```
(AshDrift) PS D:\MUJ\Projects\data-drift-prediction-in-ml-models\src> python main.py
2026-04-04 19:19:01,512 | INFO | === Training Pipeline Started ===
2026-04-04 19:19:01,514 | INFO | Processed data not found. Running preprocessing...
2026-04-04 19:19:01,515 | INFO | === Stage 1: Streaming raw files to interim/ ===
2026-04-04 19:19:01,516 | INFO | Streaming: Monday-WorkingHours.pcap_ISCX.csv
2026-04-04 19:19:19,322 | INFO |   [Monday-WorkingHours.pcap_ISCX.] kept 529,481 rows, removed 437 NaN/Inf rows
2026-04-04 19:19:19,390 | INFO | Streaming: Tuesday-WorkingHours.pcap_ISCX.csv
2026-04-04 19:19:34,910 | INFO |   [Tuesday-WorkingHours.pcap_ISCX] kept 445,645 rows, removed 264 NaN/Inf rows
2026-04-04 19:19:34,969 | INFO | Streaming: Wednesday-workingHours.pcap_ISCX.csv
2026-04-04 19:19:58,849 | INFO |   [Wednesday-workingHours.pcap_IS] kept 691,406 rows, removed 1,297 NaN/Inf rows
2026-04-04 19:19:58,891 | INFO | Streaming: Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv
2026-04-04 19:20:04,369 | INFO |   [Thursday-WorkingHours-Morning-] kept 170,231 rows, removed 135 NaN/Inf rows
2026-04-04 19:20:04,411 | INFO | Streaming: Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv
2026-04-04 19:20:14,024 | INFO |   [Thursday-WorkingHours-Afternoo] kept 288,395 rows, removed 207 NaN/Inf rows
2026-04-04 19:20:14,082 | INFO | Streaming: Friday-WorkingHours-Morning.pcap_ISCX.csv
2026-04-04 19:20:20,777 | INFO |   [Friday-WorkingHours-Morning.pc] kept 190,911 rows, removed 122 NaN/Inf rows
2026-04-04 19:20:20,825 | INFO | Streaming: Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv
2026-04-04 19:20:29,485 | INFO |   [Friday-WorkingHours-Afternoon-] kept 286,096 rows, removed 371 NaN/Inf rows
2026-04-04 19:20:29,547 | INFO | Streaming: Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv
2026-04-04 19:20:36,836 | INFO |   [Friday-WorkingHours-Afternoon-] kept 225,711 rows, removed 34 NaN/Inf rows
2026-04-04 19:20:36,878 | INFO | Stage 1 complete: 2,827,876 rows written to interim/
2026-04-04 19:20:36,879 | INFO | === Stage 2: Fitting feature selector on Monday ===
2026-04-04 19:20:36,880 | INFO | Sampling reference data for feature selection
2026-04-04 19:20:43,946 | INFO | Low variance removed: 10 features
2026-04-04 19:20:45,031 | INFO | Highly correlated dropped: 21 features
```

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2026-04-04 19:20:45,031 | INFO | Features retained after selection: 46
2026-04-04 19:20:45,073 | INFO | === Stage 3: Building processed CSV ===
2026-04-04 19:20:45,075 | INFO | Processing: Monday-WorkingHours.pcap_ISCX.csv
2026-04-04 19:20:45,591 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:46,326 | INFO |   Written 50,000 rows (total so far: 50,000)
2026-04-04 19:20:46,919 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:47,695 | INFO |   Written 50,000 rows (total so far: 100,000)
2026-04-04 19:20:48,222 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:48,849 | INFO |   Written 50,000 rows (total so far: 150,000)
2026-04-04 19:20:49,473 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:50,262 | INFO |   Written 50,000 rows (total so far: 200,000)
2026-04-04 19:20:50,933 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:51,854 | INFO |   Written 50,000 rows (total so far: 250,000)
2026-04-04 19:20:52,431 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:53,355 | INFO |   Written 50,000 rows (total so far: 300,000)
2026-04-04 19:20:53,942 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:54,565 | INFO |   Written 50,000 rows (total so far: 350,000)
2026-04-04 19:20:55,064 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:55,788 | INFO |   Written 50,000 rows (total so far: 400,000)
2026-04-04 19:20:56,349 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:57,077 | INFO |   Written 50,000 rows (total so far: 450,000)
2026-04-04 19:20:57,587 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:58,330 | INFO |   Written 50,000 rows (total so far: 500,000)
2026-04-04 19:20:58,796 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:20:59,202 | INFO |   Written 29,481 rows (total so far: 529,481)
2026-04-04 19:20:59,260 | INFO | Processing: Tuesday-WorkingHours.pcap_ISCX.csv
2026-04-04 19:20:59,972 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:00,790 | INFO |   Written 50,000 rows (total so far: 579,481)
2026-04-04 19:21:01,465 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:02,449 | INFO |   Written 50,000 rows (total so far: 629,481)
2026-04-04 19:21:03,035 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:03,868 | INFO |   Written 50,000 rows (total so far: 679,481)
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2026-04-04 19:21:04,410 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:05,270 | INFO |   Written 50,000 rows (total so far: 729,481)
2026-04-04 19:21:05,946 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:06,890 | INFO |   Written 50,000 rows (total so far: 779,481)
2026-04-04 19:21:07,367 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:08,025 | INFO |   Written 50,000 rows (total so far: 829,481)
2026-04-04 19:21:08,503 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:09,273 | INFO |   Written 50,000 rows (total so far: 879,481)
2026-04-04 19:21:09,806 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:10,505 | INFO |   Written 50,000 rows (total so far: 929,481)
2026-04-04 19:21:10,958 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:11,499 | INFO |   Written 45,645 rows (total so far: 975,126)
2026-04-04 19:21:11,563 | INFO | Processing: Wednesday-workingHours.pcap_ISCX.csv
2026-04-04 19:21:12,174 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:12,961 | INFO |   Written 50,000 rows (total so far: 1,025,126)
2026-04-04 19:21:13,627 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:14,458 | INFO |   Written 50,000 rows (total so far: 1,075,126)
2026-04-04 19:21:15,030 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:15,736 | INFO |   Written 50,000 rows (total so far: 1,125,126)
2026-04-04 19:21:16,393 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:17,323 | INFO |   Written 50,000 rows (total so far: 1,175,126)
2026-04-04 19:21:17,867 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:18,572 | INFO |   Written 50,000 rows (total so far: 1,225,126)
2026-04-04 19:21:19,201 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:19,974 | INFO |   Written 50,000 rows (total so far: 1,275,126)
2026-04-04 19:21:20,461 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:21,321 | INFO |   Written 50,000 rows (total so far: 1,325,126)
2026-04-04 19:21:22,055 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:22,736 | INFO |   Written 50,000 rows (total so far: 1,375,126)
2026-04-04 19:21:23,233 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:23,964 | INFO |   Written 50,000 rows (total so far: 1,425,126)
2026-04-04 19:21:24,646 | INFO |   apply_feature_selector: dropped 31 columns
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2026-04-04 19:21:25,444 | INFO |   Written 50,000 rows (total so far: 1,475,126)
2026-04-04 19:21:26,069 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:26,773 | INFO |   Written 50,000 rows (total so far: 1,525,126)
2026-04-04 19:21:27,305 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:27,864 | INFO |   Written 50,000 rows (total so far: 1,575,126)
2026-04-04 19:21:28,396 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:29,032 | INFO |   Written 50,000 rows (total so far: 1,625,126)
2026-04-04 19:21:29,584 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:30,231 | INFO |   Written 41,406 rows (total so far: 1,666,532)
2026-04-04 19:21:30,290 | INFO | Processing: Thursday-WorkingHours-Morning-
WebAttacks.pcap_ISCX.csv
2026-04-04 19:21:30,827 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:31,570 | INFO |   Written 50,000 rows (total so far: 1,716,532)
2026-04-04 19:21:32,202 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:32,843 | INFO |   Written 50,000 rows (total so far: 1,766,532)
2026-04-04 19:21:33,358 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:34,024 | INFO |   Written 50,000 rows (total so far: 1,816,532)
2026-04-04 19:21:34,316 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:34,596 | INFO |   Written 20,231 rows (total so far: 1,836,763)
2026-04-04 19:21:34,631 | INFO | Processing: Thursday-WorkingHours-Afternoon-
Infilteration.pcap_ISCX.csv
2026-04-04 19:21:35,293 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:36,007 | INFO |   Written 50,000 rows (total so far: 1,886,763)
2026-04-04 19:21:36,513 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:37,244 | INFO |   Written 50,000 rows (total so far: 1,936,763)
2026-04-04 19:21:37,753 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:38,412 | INFO |   Written 50,000 rows (total so far: 1,986,763)
2026-04-04 19:21:38,873 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:39,582 | INFO |   Written 50,000 rows (total so far: 2,036,763)
2026-04-04 19:21:40,175 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:41,068 | INFO |   Written 50,000 rows (total so far: 2,086,763)
2026-04-04 19:21:41,459 | INFO |   apply_feature_selector: dropped 31 columns
2026-04-04 19:21:42,046 | INFO |   Written 38,395 rows (total so far: 2,125,158)
```

2026-04-04 19:21:42,097 | INFO | Processing: Friday-WorkingHours-Morning.pcap_ISCX.csv

2026-04-04 19:21:42,630 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:43,510 | INFO | Written 50,000 rows (total so far: 2,175,158)

2026-04-04 19:21:44,074 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:44,749 | INFO | Written 50,000 rows (total so far: 2,225,158)

2026-04-04 19:21:45,348 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:46,098 | INFO | Written 50,000 rows (total so far: 2,275,158)

2026-04-04 19:21:46,505 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:47,106 | INFO | Written 40,911 rows (total so far: 2,316,069)

2026-04-04 19:21:47,154 | INFO | Processing: Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv

2026-04-04 19:21:47,852 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:48,492 | INFO | Written 50,000 rows (total so far: 2,366,069)

2026-04-04 19:21:48,984 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:49,693 | INFO | Written 50,000 rows (total so far: 2,416,069)

2026-04-04 19:21:50,113 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:50,840 | INFO | Written 50,000 rows (total so far: 2,466,069)

2026-04-04 19:21:51,391 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:52,188 | INFO | Written 50,000 rows (total so far: 2,516,069)

2026-04-04 19:21:52,671 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:53,236 | INFO | Written 50,000 rows (total so far: 2,566,069)

2026-04-04 19:21:53,653 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:54,184 | INFO | Written 36,096 rows (total so far: 2,602,165)

2026-04-04 19:21:54,229 | INFO | Processing: Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv

2026-04-04 19:21:54,804 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:55,605 | INFO | Written 50,000 rows (total so far: 2,652,165)

2026-04-04 19:21:56,217 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:56,973 | INFO | Written 50,000 rows (total so far: 2,702,165)

2026-04-04 19:21:57,513 | INFO | apply_feature_selector: dropped 31 columns

2026-04-04 19:21:58,353 | INFO | Written 50,000 rows (total so far: 2,752,165)

2026-04-04 19:21:58,943 | INFO | apply_feature_selector: dropped 31 columns

```
2026-04-04 19:21:59,617 | INFO |   Written 50,000 rows (total so far: 2,802,165)

2026-04-04 19:21:59,901 | INFO |   apply_feature_selector: dropped 31 columns

2026-04-04 19:22:00,242 | INFO |   Written 25,711 rows (total so far: 2,827,876)

2026-04-04 19:22:00,290 | INFO | Stage 3 complete: 2,827,876 rows in processed CSV

2026-04-04 19:22:00,290 | INFO | === Stage 4: Scanning for globally constant columns ===

2026-04-04 19:22:25,814 | INFO | No constant columns found across combined dataset.

2026-04-04 19:22:25,814 | INFO | Loading processed dataset: D:\MUJ\Projects\data-drift-prediction-in-ml-models\data\processed\processed_data_v1.csv

2026-04-04 19:22:25,816 | INFO | DEBUG mode: Chunk-loading a sample to prevent memory crash...

2026-04-04 19:22:41,610 | INFO | Loaded 100,000 rows | Memory: 47.4 MB

2026-04-04 19:22:41,834 | INFO | Train: 80000 | Test: 20000 | Features: 46

2026-04-04 19:22:41,983 | INFO | Scaler saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\scalers\scaler_v1.pkl

2026-04-04 19:22:41,984 | INFO |

=== Training: random_forest ===

[Parallel(n_jobs=-1)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=-1)]: Done  26 tasks      | elapsed:    1.5s

[Parallel(n_jobs=-1)]: Done 100 out of 100 | elapsed:    4.5s finished

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done  26 tasks      | elapsed:    0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed:    0.0s finished

2026-04-04 19:22:47,060 | INFO | Accuracy: 0.9976

2026-04-04 19:22:47,060 | INFO |

              precision    recall  f1-score   support


           0       1.00      1.00      1.00     16057

           1       1.00      0.99      0.99      3943


    accuracy                           1.00     20000

   macro avg       1.00      1.00      1.00     20000

weighted avg       1.00      1.00      1.00     20000
```

2026-04-04 19:22:47,072 | INFO | Confusion Matrix:

[[16040 17]

 [32 3911]]

2026-04-04 19:22:47,289 | INFO | Model saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\model\random_forest_v1.pkl

Report saved → D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\experiment_20260404_192247.txt

2026-04-04 19:22:47,297 | INFO |

=== Training: xgboost ===

2026-04-04 19:22:49,324 | INFO | Accuracy: 0.9979

2026-04-04 19:22:49,325 | INFO |

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     16057
           1       1.00      0.99      0.99      3943

    accuracy                           1.00     20000
   macro avg       1.00      1.00      1.00     20000
weighted avg       1.00      1.00      1.00     20000
```

2026-04-04 19:22:49,326 | INFO | Confusion Matrix:

[[16040 17]

 [26 3917]]

2026-04-04 19:22:49,355 | INFO | Model saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\model\xgboost_v1.pkl

Report saved → D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\experiment_20260404_192249.txt

2026-04-04 19:22:49,360 | INFO |

=== Training: logistic ===

D:\MUJ\Projects\data-drift-prediction-in-ml-models\AshDrift\Lib\site-packages\sklearn\linear_model_logistic.py:1184: FutureWarning: 'n_jobs' has no effect since 1.8 and will be removed in 1.10. You provided 'n_jobs=-1', please leave it unspecified.

  warnings.warn(msg, category=FutureWarning)

2026-04-04 19:22:51,162 | INFO | Accuracy: 0.9383

2026-04-04 19:22:51,162 | INFO |

```
              precision    recall  f1-score   support

           0       0.97      0.96      0.96     16057
           1       0.83      0.86      0.85      3943

    accuracy                           0.94     20000
   macro avg       0.90      0.91      0.90     20000
weighted avg       0.94      0.94      0.94     20000
```

2026-04-04 19:22:51,164 | INFO | Confusion Matrix:

```
[[15380   677]
 [  557  3386]]
```

2026-04-04 19:22:51,196 | INFO | Model saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\model\logistic_v1.pkl

Report saved → D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\experiment_20260404_192251.txt

2026-04-04 19:22:51,198 | INFO |

=== Training: decision_tree ===

2026-04-04 19:22:53,331 | INFO | Accuracy: 0.9929

2026-04-04 19:22:53,332 | INFO |

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     16057
           1       0.98      0.98      0.98      3943

    accuracy                           0.99     20000
   macro avg       0.99      0.99      0.99     20000
weighted avg       0.99      0.99      0.99     20000
```

2026-04-04 19:22:53,333 | INFO | Confusion Matrix:

```
[[15982    75]
 [   66  3877]]
```

2026-04-04 19:22:53,348 | INFO | Model saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\model\decision_tree_v1.pkl

Report saved → D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\experiment_20260404_192253.txt

2026-04-04 19:22:53,351 | INFO |

=== Training: svm ===

2026-04-04 19:23:08,079 | INFO | Accuracy: 0.9359

2026-04-04 19:23:08,080 | INFO |

```
              precision    recall  f1-score   support

           0       0.96      0.96      0.96     16057
           1       0.83      0.85      0.84      3943

    accuracy                           0.94     20000
   macro avg       0.90      0.90      0.90     20000
weighted avg       0.94      0.94      0.94     20000
```

2026-04-04 19:23:08,081 | INFO | Confusion Matrix:

```
[[15378   679]
 [  602  3341]]
```

2026-04-04 19:23:08,109 | INFO | Model saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\model\svm_v1.pkl

Report saved → D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\experiment_20260404_192308.txt

(AshDrift) PS D:\MUJ\Projects\data-drift-prediction-in-ml-models\src> python testdrift.py

2026-04-04 19:29:44,294 | INFO | === Drift Experiment Started ===

2026-04-04 19:29:44,299 | INFO | Loading: D:\MUJ\Projects\data-drift-prediction-in-ml-models\data\processed\processed_data_v1.csv

2026-04-04 19:29:44,299 | INFO | DEBUG mode: Chunk-loading to prevent memory crash...

2026-04-04 19:30:06,634 | INFO | Total rows loaded: 282788

2026-04-04 19:30:06,700 | INFO | Batch [Monday-WorkingHours.pcap_ISCX.csv]: 52939 rows

2026-04-04 19:30:06,733 | INFO | Batch [Tuesday-WorkingHours.pcap_ISCX.csv]: 44544 rows

2026-04-04 19:30:06,790 | INFO | Batch [Wednesday-workingHours.pcap_ISCX.csv]: 69134 rows

2026-04-04 19:30:06,816 | INFO | Batch [Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv]: 17074 rows

2026-04-04 19:30:06,845 | INFO | Batch [Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv]: 28795 rows

2026-04-04 19:30:06,867 | INFO | Batch [Friday-WorkingHours-Morning.pcap_ISCX.csv]: 19088 rows

2026-04-04 19:30:06,895 | INFO | Batch [Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv]: 28623 rows

2026-04-04 19:30:06,916 | INFO | Batch [Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv]: 22591 rows

2026-04-04 19:30:06,916 | INFO | Total batches found: 8

2026-04-04 19:30:06,918 | INFO |

--- Label Distribution per Batch ---

2026-04-04 19:30:06,942 | INFO | [Monday-WorkingHours.pcap_ISCX.csv] Label distribution:

Label

0 1.0

Name: proportion, dtype: float64

2026-04-04 19:30:06,945 | INFO | [Tuesday-WorkingHours.pcap_ISCX.csv] Label distribution:

Label

0 0.96893

1 0.03107

Name: proportion, dtype: float64

2026-04-04 19:30:06,947 | INFO | [Wednesday-workingHours.pcap_ISCX.csv] Label distribution:

Label

0 0.634825

1 0.365175

Name: proportion, dtype: float64

2026-04-04 19:30:06,948 | INFO | [Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv] Label distribution:

Label

0 0.988228

1 0.011772

Name: proportion, dtype: float64

2026-04-04 19:30:06,950 | INFO | [Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv] Label distribution:

Label

0 0.999861

1 0.000139

Name: proportion, dtype: float64

2026-04-04 19:30:06,951 | INFO | [Friday-WorkingHours-Morning.pcap_ISCX.csv] Label distribution:

Label

0 0.989575

1 0.010425

Name: proportion, dtype: float64

2026-04-04 19:30:06,956 | INFO | [Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv] Label distribution:

Label

0 0.445201

1 0.554799

Name: proportion, dtype: float64

2026-04-04 19:30:06,958 | INFO | [Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv] Label distribution:

Label

0 0.43327

1 0.56673

Name: proportion, dtype: float64

2026-04-04 19:30:06,958 | INFO |

--- Fixed Reference Drift (Monday vs all batches) ---

2026-04-04 19:30:07,603 | INFO | [Fixed] 0 vs 1 | KL=0.5770 | JS=0.0109 | WS=58459.0496 | FeatJS=0.0006 | Tuesday-WorkingHours.pcap_ISCX.csv

2026-04-04 19:30:08,555 | INFO | [Fixed] 0 vs 2 | KL=7.7521 | JS=0.1471 | WS=1841756.3783 | FeatJS=0.0235 | Wednesday-workingHours.pcap_ISCX.csv

2026-04-04 19:30:08,998 | INFO | [Fixed] 0 vs 3 | KL=0.2071 | JS=0.0041 | WS=108933.2041 | FeatJS=0.0013 | Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv

2026-04-04 19:30:09,576 | INFO | [Fixed] 0 vs 4 | KL=0.0018 | JS=0.0000 | WS=178856.0537 | FeatJS=0.0027 | Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv

2026-04-04 19:30:10,165 | INFO | [Fixed] 0 vs 5 | KL=0.1821 | JS=0.0036 | WS=114944.2999 | FeatJS=0.0013 | Friday-WorkingHours-Morning.pcap_ISCX.csv

2026-04-04 19:30:10,678 | INFO | [Fixed] 0 vs 6 | KL=12.0876 | JS=0.2469 | WS=390731.0981 | FeatJS=0.0110 | Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv

2026-04-04 19:30:11,269 | INFO | [Fixed] 0 vs 7 | KL=12.3652 | JS=0.2540 | WS=984674.1021 | FeatJS=0.0341 | Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv

2026-04-04 19:30:11,275 | INFO | Drift results saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\drift_scores.json

2026-04-04 19:30:11,276 | INFO |

--- Rolling Reference Window Drift ---

2026-04-04 19:30:11,933 | INFO | [Rolling] 0 -> 1 | KL=0.5770 | JS=0.0109 | WS=58459.0496 | Monday-WorkingHours.pcap_ISCX.csv -> Tuesday-WorkingHours.pcap_ISCX.csv

2026-04-04 19:30:12,837 | INFO | [Rolling] 1 -> 2 | KL=0.6314 | JS=0.1004 | WS=1845590.1866 | Tuesday-WorkingHours.pcap_ISCX.csv -> Wednesday-workingHours.pcap_ISCX.csv

2026-04-04 19:30:13,559 | INFO | [Rolling] 2 -> 3 | KL=0.3969 | JS=0.1238 | WS=1811922.2259 | Wednesday-workingHours.pcap_ISCX.csv -> Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv

2026-04-04 19:30:13,902 | INFO | [Rolling] 3 -> 4 | KL=0.0111 | JS=0.0038 | WS=196961.0338 | Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv -> Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv

2026-04-04 19:30:14,365 | INFO | [Rolling] 4 -> 5 | KL=0.0348 | JS=0.0033 | WS=259451.7737 | Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv -> Friday-WorkingHours-Morning.pcap_ISCX.csv

2026-04-04 19:30:14,747 | INFO | [Rolling] 5 -> 6 | KL=1.8494 | JS=0.2229 | WS=479789.5165 | Friday-WorkingHours-Morning.pcap_ISCX.csv -> Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv

2026-04-04 19:30:15,142 | INFO | [Rolling] 6 -> 7 | KL=0.0003 | JS=0.0001 | WS=1215287.8146 | Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv -> Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv

2026-04-04 19:30:15,146 | INFO | Rolling drift results saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\rolling_drift_scores.json

2026-04-04 19:30:15,147 | INFO |

--- Multi-Model Cross-Dataset Evaluation ---

Models: ['random_forest', 'xgboost', 'logistic', 'decision_tree', 'svm']

2026-04-04 19:30:16,705 | INFO |

--- Cross-eval model: random_forest ---

[Parallel(n_jobs=-1)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=-1)]: Done 26 tasks | elapsed: 0.2s

[Parallel(n_jobs=-1)]: Done 100 out of 100 | elapsed: 0.4s finished

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.0s finished

2026-04-04 19:30:17,318 | INFO | Self-accuracy: 1.0000

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.0s finished

2026-04-04 19:30:17,394 | INFO | Batch 1 [Tuesday-WorkingHours.pcap_ISCX.csv]: 0.9689

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.0s finished

2026-04-04 19:30:17,470 | INFO | Batch 2 [Wednesday-workingHours.pcap_ISCX.cs]: 0.6348

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.0s finished

2026-04-04 19:30:17,550 | INFO | Batch 3 [Thursday-WorkingHours-Morning-WebAt]: 0.9882

```
[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.
[Parallel(n_jobs=12)]: Done  26 tasks      | elapsed:    0.0s
[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed:    0.0s finished
2026-04-04 19:30:17,633 | INFO |   Batch 4 [Thursday-WorkingHours-Afternoon-Inf]: 0.9999
[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.
[Parallel(n_jobs=12)]: Done  26 tasks      | elapsed:    0.1s
[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed:    0.1s finished
2026-04-04 19:30:17,901 | INFO |   Batch 5 [Friday-WorkingHours-Morning.pcap_IS]: 0.9896
[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.
[Parallel(n_jobs=12)]: Done  26 tasks      | elapsed:    0.0s
[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed:    0.0s finished
2026-04-04 19:30:17,983 | INFO |   Batch 6 [Friday-WorkingHours-Afternoon-PortS]: 0.4452
[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.
[Parallel(n_jobs=12)]: Done  26 tasks      | elapsed:    0.0s
[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed:    0.0s finished
2026-04-04 19:30:18,133 | INFO |   Batch 7 [Friday-WorkingHours-Afternoon-DDos.]: 0.4333
2026-04-04 19:30:18,134 | INFO |
--- Cross-eval model: xgboost ---
2026-04-04 19:30:18,812 | INFO |   Self-accuracy: 1.0000
2026-04-04 19:30:18,863 | INFO |   Batch 1 [Tuesday-WorkingHours.pcap_ISCX.csv]: 0.9689
2026-04-04 19:30:18,936 | INFO |   Batch 2 [Wednesday-workingHours.pcap_ISCX.cs]: 0.6348
2026-04-04 19:30:18,965 | INFO |   Batch 3 [Thursday-WorkingHours-Morning-WebAt]: 0.9882
2026-04-04 19:30:19,001 | INFO |   Batch 4 [Thursday-WorkingHours-Afternoon-Inf]: 0.9999
2026-04-04 19:30:19,027 | INFO |   Batch 5 [Friday-WorkingHours-Morning.pcap_IS]: 0.9896
2026-04-04 19:30:19,063 | INFO |   Batch 6 [Friday-WorkingHours-Afternoon-PortS]: 0.4452
2026-04-04 19:30:19,095 | INFO |   Batch 7 [Friday-WorkingHours-Afternoon-DDos.]: 0.4333
2026-04-04 19:30:19,096 | INFO |
--- Cross-eval model: logistic ---
D:\MUJ\Projects\data-drift-prediction-in-ml-models\AshDrift\Lib\site-packages\sklearn\linear_model\_logistic.py:1184: FutureWarning: 'n_jobs' has no effect since 1.8 and will be removed in 1.10. You provided 'n_jobs=-1', please leave it unspecified.
```

```
  warnings.warn(msg, category=FutureWarning)
Traceback (most recent call last):
  File "D:\MUJ\Projects\data-drift-prediction-in-ml-models\src\testdrift.py", line 144, in <module>
    run_drift_experiment()
    ~~~~~~~~~~~~~~~~~~~~^^
    ~~~~~~~~~~~~~~~~~~~~^^
  File "D:\MUJ\Projects\data-drift-prediction-in-ml-models\src\testdrift.py", line 98, in run_drift_experiment
    eval_results = cross_dataset_evaluation(batches, model_types=CROSS_EVAL_MODELS)
    ~~~~~~~~~~~~~~~~~~~~^^
    ~~~~~~~~~~~~~~~~~~~~^^
  File "D:\MUJ\Projects\data-drift-prediction-in-ml-models\src\testdrift.py", line 98, in run_drift_experiment
    eval_results = cross_dataset_evaluation(batches, model_types=CROSS_EVAL_MODELS)
  File "D:\MUJ\Projects\data-drift-prediction-in-ml-models\src\drift.py", line 342, in cross_dataset_evaluation
    model.fit(X_train, y_ref)
    ~~~~~~~~~~^^^^^^^^^^^^^^^
  File "D:\MUJ\Projects\data-drift-prediction-in-ml-models\AshDrift\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "D:\MUJ\Projects\data-drift-prediction-in-ml-models\AshDrift\Lib\site-packages\sklearn\linear_model\_logistic.py", line 1243, in fit
    raise ValueError(
    ...<3 lines>...
    )
ValueError: This solver needs samples of at least 2 classes in the data, but the data contains only one class: np.int64(0)
```

```
(AshDrift) PS D:\MUJ\Projects\data-drift-prediction-in-ml-models\src> python testdrift.py

2026-04-04 21:13:03,698 | INFO | === Drift Experiment Started ===

2026-04-04 21:13:03,700 | INFO | Loading: D:\MUJ\Projects\data-drift-prediction-in-ml-models\data\processed\processed_data_v1.csv

2026-04-04 21:13:03,700 | INFO | DEBUG mode: Chunk-loading to prevent memory crash...

2026-04-04 21:13:25,251 | INFO | Total rows loaded: 282788

2026-04-04 21:13:25,321 | INFO | Batch [Monday-WorkingHours.pcap_ISCX.csv]: 52939 rows

2026-04-04 21:13:25,353 | INFO | Batch [Tuesday-WorkingHours.pcap_ISCX.csv]: 44544 rows

2026-04-04 21:13:25,391 | INFO | Batch [Wednesday-workingHours.pcap_ISCX.csv]: 69134 rows

2026-04-04 21:13:25,413 | INFO | Batch [Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv]: 17074 rows

2026-04-04 21:13:25,440 | INFO | Batch [Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv]: 28795 rows

2026-04-04 21:13:25,464 | INFO | Batch [Friday-WorkingHours-Morning.pcap_ISCX.csv]: 19088 rows

2026-04-04 21:13:25,496 | INFO | Batch [Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv]: 28623 rows

2026-04-04 21:13:25,525 | INFO | Batch [Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv]: 22591 rows

2026-04-04 21:13:25,526 | INFO | Total batches found: 8

2026-04-04 21:13:25,527 | INFO |

--- Label Distribution per Batch ---

2026-04-04 21:13:25,640 | INFO | [Monday-WorkingHours.pcap_ISCX.csv] Label distribution:

Label

0    1.0

Name: proportion, dtype: float64


2026-04-04 21:13:25,646 | INFO | [Tuesday-WorkingHours.pcap_ISCX.csv] Label distribution:

Label

0    0.96893

1    0.03107
```

Name: proportion, dtype: float64

2026-04-04 21:13:25,655 | INFO | [Wednesday-workingHours.pcap_ISCX.csv] Label distribution:

Label

0 0.634825

1 0.365175

Name: proportion, dtype: float64

2026-04-04 21:13:25,661 | INFO | [Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv] Label distribution:

Label

0 0.988228

1 0.011772

Name: proportion, dtype: float64

2026-04-04 21:13:25,666 | INFO | [Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv] Label distribution:

Label

0 0.999861

1 0.000139

Name: proportion, dtype: float64

2026-04-04 21:13:25,669 | INFO | [Friday-WorkingHours-Morning.pcap_ISCX.csv] Label distribution:

Label

0 0.989575

1 0.010425

Name: proportion, dtype: float64

2026-04-04 21:13:25,683 | INFO | [Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv] Label distribution:

Label

0 0.445201

1 0.554799

Name: proportion, dtype: float64

2026-04-04 21:13:25,703 | INFO | [Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv] Label distribution:

Label

0 0.43327

1 0.56673

Name: proportion, dtype: float64

2026-04-04 21:13:25,705 | INFO |

--- Fixed Reference Drift (Monday vs all batches) ---

2026-04-04 21:13:26,759 | INFO | [Fixed] 0 vs 1 | KL=0.5770 | JS=0.0109 | WS=58459.0496 | FeatJS=0.0006 | Tuesday-WorkingHours.pcap_ISCX.csv

2026-04-04 21:13:27,750 | INFO | [Fixed] 0 vs 2 | KL=7.7521 | JS=0.1471 | WS=1841756.3783 | FeatJS=0.0235 | Wednesday-workingHours.pcap_ISCX.csv

2026-04-04 21:13:28,263 | INFO | [Fixed] 0 vs 3 | KL=0.2071 | JS=0.0041 | WS=108933.2041 | FeatJS=0.0013 | Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv

2026-04-04 21:13:28,788 | INFO | [Fixed] 0 vs 4 | KL=0.0018 | JS=0.0000 | WS=178856.0537 | FeatJS=0.0027 | Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv

2026-04-04 21:13:29,356 | INFO | [Fixed] 0 vs 5 | KL=0.1821 | JS=0.0036 | WS=114944.2999 | FeatJS=0.0013 | Friday-WorkingHours-Morning.pcap_ISCX.csv

2026-04-04 21:13:29,953 | INFO | [Fixed] 0 vs 6 | KL=12.0876 | JS=0.2469 | WS=390731.0981 | FeatJS=0.0110 | Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv

2026-04-04 21:13:30,640 | INFO | [Fixed] 0 vs 7 | KL=12.3652 | JS=0.2540 | WS=984674.1021 | FeatJS=0.0341 | Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv

2026-04-04 21:13:30,668 | INFO | Drift results saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\drift_scores.json

2026-04-04 21:13:30,669 | INFO |

--- Rolling Reference Window Drift ---

2026-04-04 21:13:31,536 | INFO | [Rolling] 0 -> 1 | KL=0.5770 | JS=0.0109 | WS=58459.0496 | Monday-WorkingHours.pcap_ISCX.csv -> Tuesday-WorkingHours.pcap_ISCX.csv

2026-04-04 21:13:32,903 | INFO | [Rolling] 1 -> 2 | KL=0.6314 | JS=0.1004 | WS=1845590.1866 | Tuesday-WorkingHours.pcap_ISCX.csv -> Wednesday-workingHours.pcap_ISCX.csv

2026-04-04 21:13:33,698 | INFO | [Rolling] 2 -> 3 | KL=0.3969 | JS=0.1238 | WS=1811922.2259 | Wednesday-workingHours.pcap_ISCX.csv -> Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv

2026-04-04 21:13:34,094 | INFO | [Rolling] 3 -> 4 | KL=0.0111 | JS=0.0038 | WS=196961.0338 | Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv -> Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv

2026-04-04 21:13:34,600 | INFO | [Rolling] 4 -> 5 | KL=0.0348 | JS=0.0033 | WS=259451.7737 | Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv -> Friday-WorkingHours-Morning.pcap_ISCX.csv

2026-04-04 21:13:34,957 | INFO | [Rolling] 5 -> 6 | KL=1.8494 | JS=0.2229 | WS=479789.5165 | Friday-WorkingHours-Morning.pcap_ISCX.csv -> Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv

2026-04-04 21:13:35,361 | INFO | [Rolling] 6 -> 7 | KL=0.0003 | JS=0.0001 | WS=1215287.8146 | Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv -> Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv

2026-04-04 21:13:35,363 | INFO | Rolling drift results saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\rolling_drift_scores.json

2026-04-04 21:13:35,364 | INFO |

--- Multi-Model Cross-Dataset Evaluation ---

Models: ['random_forest', 'xgboost', 'logistic', 'decision_tree', 'svm']

2026-04-04 21:13:37,311 | WARNING | Reference batch 'Monday-WorkingHours.pcap_ISCX.csv' contains only 1 class(es) — solvers that require 2+ classes (logistic, svm) will use a DummyClassifier fallback.

2026-04-04 21:13:37,314 | INFO |

--- Cross-eval model: random_forest ---

[Parallel(n_jobs=-1)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=-1)]: Done 26 tasks | elapsed: 0.4s

[Parallel(n_jobs=-1)]: Done 100 out of 100 | elapsed: 0.8s finished

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.0s finished

2026-04-04 21:13:38,382 | INFO | Self-accuracy: 1.0000

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.0s finished

2026-04-04 21:13:38,578 | INFO | Batch 1 [Tuesday-WorkingHours.pcap_ISCX.csv]: 0.9689

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.1s finished

2026-04-04 21:13:38,939 | INFO | Batch 2 [Wednesday-workingHours.pcap_ISCX.cs]: 0.6348

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.0s finished

2026-04-04 21:13:39,106 | INFO | Batch 3 [Thursday-WorkingHours-Morning-WebAt]: 0.9882

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.0s finished

2026-04-04 21:13:39,248 | INFO | Batch 4 [Thursday-WorkingHours-Afternoon-Inf]: 0.9999

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.3s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.4s finished

2026-04-04 21:13:39,808 | INFO | Batch 5 [Friday-WorkingHours-Morning.pcap_IS]: 0.9896

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.0s finished

2026-04-04 21:13:39,962 | INFO | Batch 6 [Friday-WorkingHours-Afternoon-PortS]: 0.4452

[Parallel(n_jobs=12)]: Using backend ThreadingBackend with 12 concurrent workers.

[Parallel(n_jobs=12)]: Done 26 tasks | elapsed: 0.0s

[Parallel(n_jobs=12)]: Done 100 out of 100 | elapsed: 0.1s finished

2026-04-04 21:13:40,171 | INFO | Batch 7 [Friday-WorkingHours-Afternoon-DDos.]: 0.4333

2026-04-04 21:13:40,172 | INFO |

--- Cross-eval model: xgboost ---

2026-04-04 21:13:41,060 | INFO | Self-accuracy: 1.0000

2026-04-04 21:13:41,097 | INFO | Batch 1 [Tuesday-WorkingHours.pcap_ISCX.csv]: 0.9689

2026-04-04 21:13:41,149 | INFO | Batch 2 [Wednesday-workingHours.pcap_ISCX.cs]: 0.6348

2026-04-04 21:13:41,170 | INFO | Batch 3 [Thursday-WorkingHours-Morning-WebAt]: 0.9882

2026-04-04 21:13:41,203 | INFO | Batch 4 [Thursday-WorkingHours-Afternoon-Inf]: 0.9999

2026-04-04 21:13:41,228 | INFO | Batch 5 [Friday-WorkingHours-Morning.pcap_IS]: 0.9896

2026-04-04 21:13:41,258 | INFO | Batch 6 [Friday-WorkingHours-Afternoon-PortS]: 0.4452

2026-04-04 21:13:41,289 | INFO | Batch 7 [Friday-WorkingHours-Afternoon-DDos.]: 0.4333

2026-04-04 21:13:41,289 | INFO |

--- Cross-eval model: logistic ---

2026-04-04 21:13:41,298 | WARNING | [logistic] Reference is single-class — using DummyClassifier (always predicts majority class).

2026-04-04 21:13:41,306 | INFO | Self-accuracy: 1.0000

2026-04-04 21:13:41,375 | INFO | Batch 1 [Tuesday-WorkingHours.pcap_ISCX.csv]: 0.9689

2026-04-04 21:13:41,458 | INFO | Batch 2 [Wednesday-workingHours.pcap_ISCX.cs]: 0.6348

2026-04-04 21:13:41,492 | INFO | Batch 3 [Thursday-WorkingHours-Morning-WebAt]: 0.9882

2026-04-04 21:13:41,547 | INFO | Batch 4 [Thursday-WorkingHours-Afternoon-Inf]: 0.9999

2026-04-04 21:13:41,575 | INFO | Batch 5 [Friday-WorkingHours-Morning.pcap_IS]: 0.9896

2026-04-04 21:13:41,632 | INFO | Batch 6 [Friday-WorkingHours-Afternoon-PortS]: 0.4452

2026-04-04 21:13:41,680 | INFO | Batch 7 [Friday-WorkingHours-Afternoon-DDos.]: 0.4333

2026-04-04 21:13:41,681 | INFO |

--- Cross-eval model: decision_tree ---

2026-04-04 21:13:41,774 | INFO | Self-accuracy: 1.0000

2026-04-04 21:13:41,839 | INFO | Batch 1 [Tuesday-WorkingHours.pcap_ISCX.csv]: 0.9689

2026-04-04 21:13:41,934 | INFO | Batch 2 [Wednesday-workingHours.pcap_ISCX.cs]: 0.6348

2026-04-04 21:13:41,980 | INFO | Batch 3 [Thursday-WorkingHours-Morning-WebAt]: 0.9882

2026-04-04 21:13:42,030 | INFO | Batch 4 [Thursday-WorkingHours-Afternoon-Inf]: 0.9999

2026-04-04 21:13:42,087 | INFO | Batch 5 [Friday-WorkingHours-Morning.pcap_IS]: 0.9896

2026-04-04 21:13:42,140 | INFO | Batch 6 [Friday-WorkingHours-Afternoon-PortS]: 0.4452

2026-04-04 21:13:42,190 | INFO | Batch 7 [Friday-WorkingHours-Afternoon-DDos.]: 0.4333

2026-04-04 21:13:42,191 | INFO |

--- Cross-eval model: svm ---

2026-04-04 21:13:42,200 | WARNING | [svm] Reference is single-class — using DummyClassifier (always predicts majority class).

2026-04-04 21:13:42,223 | INFO | Self-accuracy: 1.0000

2026-04-04 21:13:42,365 | INFO | Batch 1 [Tuesday-WorkingHours.pcap_ISCX.csv]: 0.9689

2026-04-04 21:13:42,486 | INFO | Batch 2 [Wednesday-workingHours.pcap_ISCX.cs]: 0.6348

2026-04-04 21:13:42,528 | INFO | Batch 3 [Thursday-WorkingHours-Morning-WebAt]: 0.9882

2026-04-04 21:13:42,584 | INFO | Batch 4 [Thursday-WorkingHours-Afternoon-Inf]: 0.9999

2026-04-04 21:13:42,626 | INFO | Batch 5 [Friday-WorkingHours-Morning.pcap_IS]: 0.9896

2026-04-04 21:13:42,691 | INFO | Batch 6 [Friday-WorkingHours-Afternoon-PortS]: 0.4452

2026-04-04 21:13:42,784 | INFO | Batch 7 [Friday-WorkingHours-Afternoon-DDos.]: 0.4333

2026-04-04 21:13:42,803 | INFO | Cross-eval results saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\cross_eval_accuracy.json

2026-04-04 21:13:42,804 | INFO |

--- Checking Drift Alerts ---

2026-04-04 21:13:42,809 | WARNING | DRIFT ALERT | Wednesday-workingHours.pcap_ISCX.csv | avg_acc=0.6348 < 0.8

2026-04-04 21:13:42,811 | WARNING | DRIFT ALERT | Friday-WorkingHours-Afternoon-PortScan.p | avg_acc=0.4452 < 0.8

2026-04-04 21:13:42,814 | WARNING | DRIFT ALERT | Friday-WorkingHours-Afternoon-DDos.pcap_ | avg_acc=0.4333 < 0.8

2026-04-04 21:13:42,823 | INFO | Alerts saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\drift_alerts.json

2026-04-04 21:13:42,825 | INFO |

=== CROSS-DATASET EVALUATION SUMMARY ===

2026-04-04 21:13:42,827 | INFO | Batch random_forest xgboost logisticdecision_tree svm

2026-04-04 21:13:42,829 | INFO | ----------

2026-04-04 21:13:42,830 | INFO | Monday-WorkingHours.pcap_ISCX.csv 1.0 1.0 1.0 1.0 1.0

```
2026-04-04 21:13:42,832 | INFO | Tuesday-WorkingHours.pcap_ISCX.csv
0.9689      0.9689      0.9689      0.9689      0.9689

2026-04-04 21:13:42,839 | INFO | Wednesday-workingHours.pcap_ISCX.csv
0.6348      0.6348      0.6348      0.6348      0.6348

2026-04-04 21:13:42,841 | INFO | Thursday-WorkingHours-Morning-
WebAttacks.pcap_ISCX.csv      0.9882      0.9882      0.9882      0.9882      0.9882

2026-04-04 21:13:42,841 | INFO | Thursday-WorkingHours-Afternoon-
Infilteration.pcap_ISCX.csv      0.9999      0.9999      0.9999      0.9999      0.9999

2026-04-04 21:13:42,841 | INFO | Friday-WorkingHours-Morning.pcap_ISCX.csv
0.9896      0.9896      0.9896      0.9896      0.9896

2026-04-04 21:13:42,842 | INFO | Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv
0.4452      0.4452      0.4452      0.4452      0.4452

2026-04-04 21:13:42,842 | INFO | Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv
0.4333      0.4333      0.4333      0.4333      0.4333

2026-04-04 21:13:42,842 | INFO |

Drift scores         -> artifacts/reports/drift_scores.json

2026-04-04 21:13:42,843 | INFO | Rolling drift        ->
artifacts/reports/rolling_drift_scores.json

2026-04-04 21:13:42,844 | INFO | Cross-eval accuracy  ->
artifacts/reports/cross_eval_accuracy.json

2026-04-04 21:13:42,845 | INFO | Drift alerts         -> artifacts/reports/drift_alerts.json  (3
alerts)

2026-04-04 21:13:42,846 | INFO | === Drift Experiment Complete ===

(AshDrift) PS D:\MUJ\Projects\data-drift-prediction-in-ml-models\src>
```

```
PS D:\MUJ\Projects\data-drift-prediction-in-ml-models\src> python shap_analysis.py

2026-04-04 22:04:13,266 | INFO | === SHAP Analysis Started ===

2026-04-04 22:04:13,272 | INFO | Loading: D:\MUJ\Projects\data-drift-prediction-in-ml-
models\data\processed\processed_data_v1.csv

2026-04-04 22:04:13,272 | INFO | DEBUG mode: Chunk-loading to prevent memory crash...

2026-04-04 22:04:40,028 | INFO | Reference : Monday-WorkingHours.pcap_ISCX.csv
(52939 rows)
```

2026-04-04 22:04:40,038 | INFO | Drifted : Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv (22591 rows)

2026-04-04 22:04:49,740 | INFO |

--- SHAP: random_forest ---

2026-04-04 22:05:20,237 | INFO | Plot saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_comparison_random_forest.png

2026-04-04 22:05:20,240 | INFO | JSON saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_values_random_forest.json

2026-04-04 22:05:20,240 | INFO | Top 5 features — reference vs drifted importance:

2026-04-04 22:05:20,242 | INFO | Destination Port ref=0.0265 drift=0.0481 delta=+0.0216

2026-04-04 22:05:20,242 | INFO | Fwd Packet Length Max ref=0.0206 drift=0.0444 delta=+0.0238

2026-04-04 22:05:20,242 | INFO | Packet Length Mean ref=0.0154 drift=0.0315 delta=+0.0161

2026-04-04 22:05:20,242 | INFO | Bwd Packet Length Std ref=0.0151 drift=0.0709 delta=+0.0558

2026-04-04 22:05:20,242 | INFO | Bwd Packet Length Mean ref=0.0132 drift=0.0556 delta=+0.0424

2026-04-04 22:05:20,242 | INFO |

--- SHAP: xgboost ---

2026-04-04 22:05:21,300 | INFO | Plot saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_comparison_xgboost.png

2026-04-04 22:05:21,302 | INFO | JSON saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_values_xgboost.json

2026-04-04 22:05:21,302 | INFO | Top 5 features — reference vs drifted importance:

2026-04-04 22:05:21,302 | INFO | Init_Win_bytes_forward ref=1.5721 drift=0.7364 delta=-0.8358

2026-04-04 22:05:21,302 | INFO | Destination Port ref=1.2514 drift=1.4743 delta=+0.2229

2026-04-04 22:05:21,304 | INFO | Fwd Packet Length Max ref=0.9991 drift=0.9372 delta=-0.0619

2026-04-04 22:05:21,305 | INFO | Bwd Packet Length Min ref=0.7446 drift=0.3660 delta=-0.3786

2026-04-04 22:05:21,305 | INFO | Init_Win_bytes_backward ref=0.6041 drift=0.5600 delta=-0.0442

2026-04-04 22:05:21,306 | INFO |

--- SHAP: decision_tree ---

2026-04-04 22:05:22,115 | INFO | Plot saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_comparison_decision_tree.png

2026-04-04 22:05:22,119 | INFO | JSON saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_values_decision_tree.json

2026-04-04 22:05:22,120 | INFO | Top 5 features — reference vs drifted importance:

2026-04-04 22:05:22,121 | INFO | Total Length of Fwd Packets ref=0.0624 drift=0.0443 delta=-0.0181

2026-04-04 22:05:22,121 | INFO | Bwd Packet Length Std ref=0.0579 drift=0.2978 delta=+0.2399

2026-04-04 22:05:22,122 | INFO | Destination Port ref=0.0536 drift=0.0526 delta=-0.0011

2026-04-04 22:05:22,123 | INFO | Flow Bytes/s ref=0.0349 drift=0.0735 delta=+0.0386

2026-04-04 22:05:22,124 | INFO | PSH Flag Count ref=0.0289 drift=0.0162 delta=-0.0127

2026-04-04 22:05:22,126 | INFO |

--- SHAP: logistic ---

2026-04-04 22:05:23,042 | INFO | Plot saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_comparison_logistic.png

2026-04-04 22:05:23,044 | INFO | JSON saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_values_logistic.json

2026-04-04 22:05:23,045 | INFO | Top 5 features — reference vs drifted importance:

2026-04-04 22:05:23,045 | INFO | Bwd Packet Length Min ref=2.8267 drift=2.4952 delta=-0.3314

2026-04-04 22:05:23,045 | INFO | Fwd Packet Length Std ref=1.9002 drift=5.5785 delta=+3.6783

2026-04-04 22:05:23,046 | INFO | PSH Flag Count ref=1.5675 drift=1.7036 delta=+0.1361

2026-04-04 22:05:23,047 | INFO | min_seg_size_forward ref=1.2170 drift=1.2017 delta=-0.0153

2026-04-04 22:05:23,047 | INFO | ACK Flag Count ref=1.1958 drift=1.4976 delta=+0.3019

2026-04-04 22:05:23,051 | INFO |

--- SHAP: svm ---

2026-04-04 22:05:23,848 | INFO | Plot saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_comparison_svm.png

2026-04-04 22:05:23,852 | INFO | JSON saved -> D:\MUJ\Projects\data-drift-prediction-in-ml-models\artifacts\reports\shap_values_svm.json

```
2026-04-04 22:05:23,853 | INFO |   Top 5 features — reference vs drifted importance:
2026-04-04 22:05:23,853 | INFO |     Max Packet Length                    ref=1.4487
drift=8.3124  delta=+6.8637
2026-04-04 22:05:23,854 | INFO |     Bwd Packet Length Max                ref=1.0757
drift=5.6681  delta=+4.5924
2026-04-04 22:05:23,854 | INFO |     PSH Flag Count                       ref=0.7133
drift=0.7752  delta=+0.0619
2026-04-04 22:05:23,855 | INFO |     Fwd Packet Length Std                ref=0.6250
drift=1.8348  delta=+1.2098
2026-04-04 22:05:23,856 | INFO |     ACK Flag Count                       ref=0.5740
drift=0.7189  delta=+0.1449
2026-04-04 22:05:23,856 | INFO |
=== SHAP Analysis Complete ===
2026-04-04 22:05:23,857 | INFO | All outputs in: D:\MUJ\Projects\data-drift-prediction-in-ml-
models\artifacts\reports
PS D:\MUJ\Projects\data-drift-prediction-in-ml-models\src>
```